Lecture 16

Harmonic functions and the Dirichlet problem

Objectives. *A prescribed continuous voltage on the boundary of a disk determines a unique harmonic potential inside it. The Poisson kernel constructs this potential and proves its boundary behaviour. Conformal maps transfer harmonic functions between planar domains.*

By the end of this lecture you will be able to:

- Prove uniqueness for continuous-boundary harmonic extension.
- Construct the disk extension using the Poisson kernel.
- Prove uniform convergence to continuous boundary data.
- Compute a disk electrostatic potential and its electric field.

16.1 Harmonic functions and the maximum principle

A real function $u \in C^2(U)$ on an open set $U \subset \mathbb{R}^2$ is *harmonic* if its Laplacian vanishes:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

The real and imaginary parts of a holomorphic function are harmonic. Indeed, the Cauchy–Riemann equations give $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$. Holomorphic functions have continuous derivatives of every order by Theorem 10.3, so the mixed derivatives agree.

The *continuous-boundary Dirichlet problem* asks for a harmonic u on a bounded domain U that is continuous on $\overline{U}$ and equals specified continuous data on ∂U.

Theorem 16.1 (Weak maximum principle)**.** *If U is bounded and $u \in C^2(U) \cap C(\overline{U})$ is real and harmonic, then*

$$\min_{\partial U} u \le u(z) \le \max_{\partial U} u \qquad (z \in \overline{U}).$$

Consequently the continuous-boundary Dirichlet problem has at most one solution.

Proof. For $\varepsilon > 0$ put $u_\varepsilon(z) = u(z) + \varepsilon |z|^2$. This continuous function attains its maximum on the compact closure. At an interior maximum each second coordinate derivative is nonpositive. That would imply $\Delta u_\varepsilon \le 0$, whereas $\Delta u_\varepsilon = 4\varepsilon > 0$. Hence its maximum occurs on the boundary. If $|z| \le M$ on the closure, then $u(z) \le \max_{\partial U} u + \varepsilon M^2$. Let $\varepsilon \downarrow 0$ for the upper bound and apply it to $-u$ for the lower bound. The difference of two solutions is harmonic and zero on the boundary. Both bounds make the difference zero. ∎

For example, zero boundary voltage forces zero potential throughout a disk. This conclusion uses continuity up to the boundary. An interior harmonic function without that condition does not yet solve this Dirichlet problem.

16.2 The Poisson kernel

For $0 \le r < 1$, define the 2π-periodic *Poisson kernel* by

$$P_r(t) = \frac{1 - r^2}{1 - 2r\cos t + r^2}. \tag{16.1}$$

It is positive. At $r = 0$ it equals one. For $r > 0$ it is largest at $t = 0$. For example, $P_{1/2}(0) = 3$ and $P_{1/2}(\pi) = 1/3$. As r increases, the kernel concentrates near integer multiples of 2π.

The holomorphic fraction underlying this kernel is

$$\operatorname{Re}\frac{e^{is} + z}{e^{is} - z} = \frac{1 - |z|^2}{|e^{is} - z|^2} = P_r(t - s), \qquad z = re^{it} \in \mathbb{D}. \tag{16.2}$$

To verify the first equality, multiply numerator and denominator by $e^{-is} - \overline{z}$. The real part of the numerator is $1 - |z|^2$. For $|z| < 1$ the geometric series gives

$$\frac{e^{is} + z}{e^{is} - z} = 1 + 2\sum_{n=1}^{\infty} z^n e^{-ins}.$$

On $|z| \le r_0 < 1$ the series converges uniformly in both z and s. Integrate term by term over $-\pi \le s \le \pi$. Every nonconstant exponential has integral zero. Taking real parts proves

$$\int_{-\pi}^{\pi} P_r(t - s)\,ds = 2\pi. \tag{16.3}$$

This is the normalisation used throughout the lecture.

Fix $0 < \delta < \pi$. On $\delta \le |t| \le \pi$, the denominator in (16.1) is $(1 - r)^2 + 2r(1 - \cos t)$. For $r \ge 1/2$ it is at least $1 - \cos\delta > 0$. Therefore

$$\sup_{\delta \le |t| \le \pi} P_r(t) \le \frac{1 - r^2}{1 - \cos\delta} \longrightarrow 0 \quad (r \uparrow 1). \tag{16.4}$$

Positivity, fixed mass and this tail bound are precisely the properties needed for continuous boundary recovery.

16.3 The disk extension

Let $f : \partial\mathbb{D} \to \mathbb{R}$ be continuous. The compact circle makes f bounded and uniformly continuous. Define its Poisson extension by

$$u(re^{it}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(t - s) f(e^{is})\,ds. \tag{16.5}$$

Stein–Shakarchi, *Fourier Analysis*, Theorem 5.7, use circle convolution with its factor $1/(2\pi)$ included. Their $f * P_r$ is exactly the extension in (16.5). Our convention writes that factor separately from the convolution, as in § 6.2.

At zero the formula is independent of t and gives the boundary average. At other points positivity and (16.3) show that the value is a weighted average of the boundary voltages.

Theorem 16.2 (Poisson extension). *For every continuous real f on $\partial\mathbb{D}$, the function in (16.5) is harmonic on $\mathbb{D}$ and extends continuously to $\overline{\mathbb{D}}$ with boundary values f. Moreover,*

$$\sup_{t \in \mathbb{R}} |u(re^{it}) - f(e^{it})| \longrightarrow 0 \quad (r \uparrow 1).$$

It is the unique solution of the continuous-boundary disk Dirichlet problem.

Proof. Define the complex function

$$h(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{is} + z}{e^{is} - z} f(e^{is})\, ds.$$

On a closed subdisk the denominator is bounded away from zero. The uniformly convergent geometric series above can be integrated term by term. Its sum is holomorphic by § 10.6. The real-part identity gives $u = \operatorname{Re} h$, which proves harmonicity.

To prove the boundary limit, subtract $f(e^{it})$ using the mass identity. Given $\varepsilon > 0$, choose $\delta > 0$ so that boundary points of angular distance less than δ have f-values differing by less than ε. The integral over that neighbourhood contributes at most ε, since its positive normalised mass is at most one. If $M = \max |f|$, the rest contributes at most

$$2M \sup_{\delta \le |s| \le \pi} P_r(s),$$

which tends to zero by (16.4). The bounds do not depend on t. This proves uniform radial convergence.

For general approach to a boundary point ζ, let $z = re^{it} \to \zeta$. The difference from $f(\zeta)$ is bounded by $|u(re^{it}) - f(e^{it})| + |f(e^{it}) - f(\zeta)|$. The first term tends to zero uniformly and the second by continuity of f. Thus the extension is continuous on the whole closed disk. Uniqueness follows from Theorem 16.1. ∎

The boundary argument uses two choices in a fixed order. Continuity of the data first fixes the arc size, uniformly over the whole circle. The tail estimate then fixes how close the evaluation point must be to the boundary. This order is why the same radial bound works at every angle and can also control a nonradial approach.

At the centre all arcs of equal length receive equal weight. Closer to the boundary, nearby arcs receive more weight. The diagram shows this dependence for the evaluation point on the real radius.

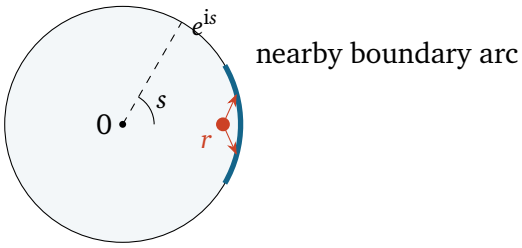


Figure 16.1: For a point r near 1, the positive Poisson weight concentrates on the boundary arc near 1. Its total normalised mass remains one.

16.4 Conformal changes of domain

A *conformal equivalence* here means a biholomorphic map between planar domains. Let $\phi : U \to V$ be such a map and let u be harmonic on V. Write $\phi = X + iY$. The real chain rule gives

$$\Delta(u \circ \phi) = u_{XX} |\nabla X|^2 + 2u_{XY} \nabla X \cdot \nabla Y + u_{YY} |\nabla Y|^2 + u_X \Delta X + u_Y \Delta Y.$$

The Cauchy–Riemann equations give $|\nabla X|^2 = |\nabla Y|^2 = |\phi'|^2$, $\nabla X \cdot \nabla Y = 0$ and $\Delta X = \Delta Y = 0$. Consequently

$$\Delta(u \circ \phi) = |\phi'|^2 (\Delta u) \circ \phi = 0. \tag{16.6}$$

This proves harmonicity of the pullback without requiring a harmonic conjugate.

A domain U is *simply connected* if every continuous closed path in U can be continuously deformed to a constant path through closed paths that remain in U. More precisely, for a closed path $\gamma : [0,1] \to U$, there are a point $z_* \in U$ and a continuous $H : [0,1]^2 \to U$ with

$$H(s,0) = \gamma(s), \qquad H(s,1) = z_*,$$
$$H(0,t) = H(1,t).$$

Only the following cited existence theorem uses this topological condition.

Theorem 16.3 (Riemann mapping · *cited*)**.** *Every nonempty simply connected proper domain $U \subsetneq \mathbb{C}$ is conformally equivalent to $\mathbb{D}$. Given $z_0 \in U$, there is a unique such map ϕ satisfying $\phi(z_0) = 0$ and $\phi'(z_0) > 0$.*

The proof of existence is external, from Stein–Shakarchi, Chapter 8, Theorem 3.1. Uniqueness follows from Theorem 15.4 after composing two maps: a disk automorphism fixing zero is a rotation, and the positive-derivative condition removes that rotation. The theorem states an interior equivalence. It does not assert that the map extends continuously to an arbitrary domain boundary. A Dirichlet solution can be transferred with specified boundary values only when the required boundary correspondence has also been justified.

16.5 A disk electrostatic potential

Consider a two-dimensional charge-free disk with boundary voltage

$$f(e^{it}) = \cos(2t) + 2\sin t.$$

The potential must be harmonic and take these boundary values. The polynomial

$$u(x,y) = x^2 - y^2 + 2y$$

has $u_{xx} + u_{yy} = 2 - 2 = 0$. On $x = \cos t$ and $y = \sin t$ its value is $\cos^2 t - \sin^2 t + 2\sin t = f(e^{it})$. Uniqueness identifies this polynomial with the Poisson extension, so

$$u(re^{it}) = r^2\cos(2t) + 2r\sin t.$$

The different angular modes acquire different radial powers. For example, $u(1/2, 1/4) = 1/4 - 1/16 + 1/2 = 11/16$.

The electric field is the negative gradient of the potential:

$$\mathbf{E}(x,y) = -\nabla u(x,y) = (-2x,\ 2y - 2).$$

Its divergence is zero in the charge-free region, and at the centre the field is $(0,-2)$. The boundary voltage determines both the interior potential and this field.

Summary.

- A harmonic function continuous on a bounded domain's closure is bounded by its boundary extrema. This proves uniqueness for the Dirichlet problem.
- The positive disk Poisson kernel has mass 2π and vanishing tails. For continuous boundary data its normalised integral is the unique harmonic extension, continuous on the closed disk.
- Conformal pullback preserves harmonicity. Riemann mapping is cited and supplies an interior disk equivalence, without a general boundary claim.

References

E. M. Stein and R. Shakarchi, *Fourier Analysis*, 2003, Chapter 2, §5, Theorem 5.7, pp. 57–58; *Complex Analysis*, 2003, Chapter 8, pp. 215–216 and Theorem 3.1, p. 224.